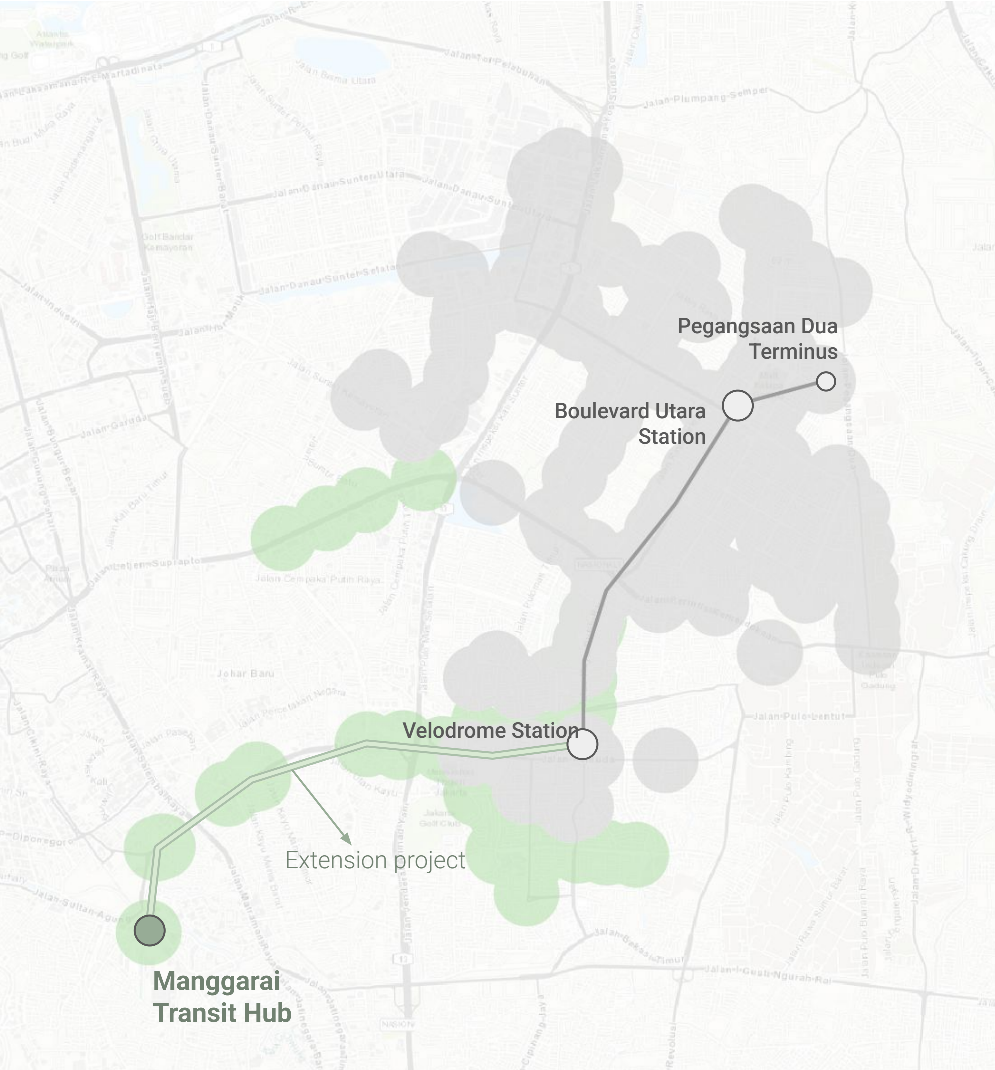
Project Sheet

Sutan Mufti
Urban And Transport Specialist

April 2026

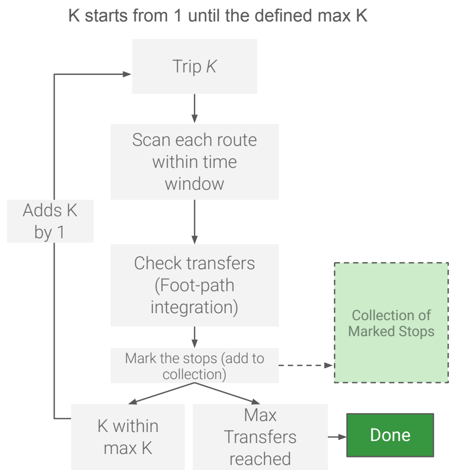
LRT Jakarta Extension Project: Boulevard Utara Catchment Simulation

Transportation Modelling
Sutan Mufti, 2026



RAPTOR Algorithm

The RAPTOR algorithm is a public transit routing method that organises calculations into discrete rounds, where each round ***K*** identifies the quickest way to reach a destination using exactly ***K*** trips. It operates directly on timetable data without using a priority queue, making it exceptionally fast for computing Pareto-optimal journeys.



This report examines the impact of the LRT Jakarta extension from Velodrome to Manggarai, focusing on the projected passenger surge at Boulevard Utara Station. By linking to the Manggarai multi-modal hub, network ridership is expected to rise from 5,000 to over 70,000 daily users with additional interventions, bringing major commercial zones into a 30-minute catchment of Central Jakarta. This enhanced connectivity unlocks previously isolated urban areas, necessitating the station's evolution into a Transit-Oriented Development (TOD) node with improved pedestrian infrastructure and intensified land-use planning.

To quantify these shifts, the RAPTOR algorithm was built to process GTFS timetable data and identify Pareto-optimal journeys across the network. The resulting data was visualised in QGIS to map the spatial expansion of 'unlocked' areas through time-isochrones. Python was used to bulk simulate several parameters. This methodology demonstrates how the extension reconfigures Boulevard Utara from a local terminal into a strategic high-capacity gateway for North Jakarta.

Tools

Go: Raptor Algorithm Implementation
QGIS: cartography

Solutions

Passenger forecast
Unlocked region identification
Development recommendations

Users

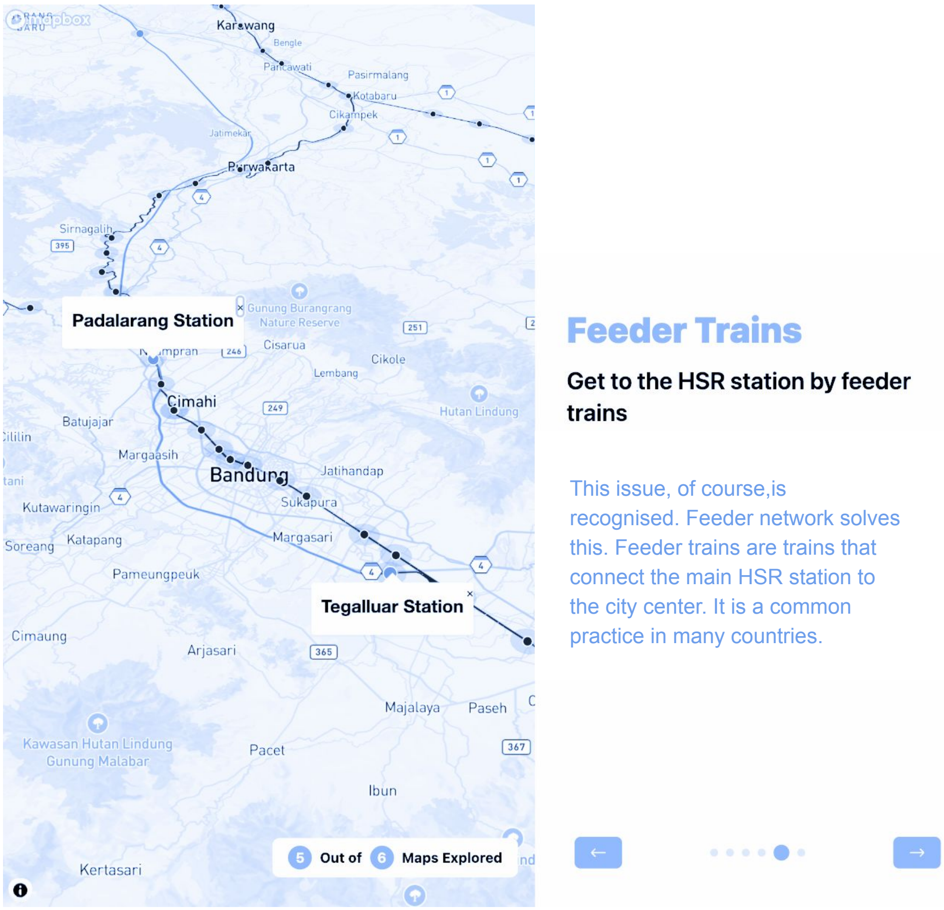
Summarecon Agung (Indonesia)
Ulab Magnaversa Indonesia

Location

Jakarta (Indonesia)

The Story of Indonesian High-Speed Rail: an Interactive Map

WebGIS 3D Visualisation
Sutan Mufti, 2025



The high-speed rail network in West Java is explored through an interactive story map hosted at ceritapeta.co.uk, which highlights the critical issue of 'out-of-place' stations situated far from city centres. This spatial disconnect is addressed through the implementation of a feeder train network, which serves as the essential link for passenger accessibility. The project showcases a high level of technical proficiency in using open-source mapping libraries and modern web frameworks to translate complex infrastructure challenges into a compelling digital narrative.

To bring this geospatial data to life, the platform was developed using SvelteKit combined with Mapbox and Maplibre. This technical stack enables the visualisation of interactive route alignments and station catchments, allowing users to engage with the data dynamically. By leveraging these tools, the story map demonstrates how advanced cartographic techniques can be used to explain the vital role of integrated transport systems in overcoming the geographical limitations of major infrastructure projects.

Tools

Typescript
Sveltekit

Solutions

Interactive map visualisation

Users

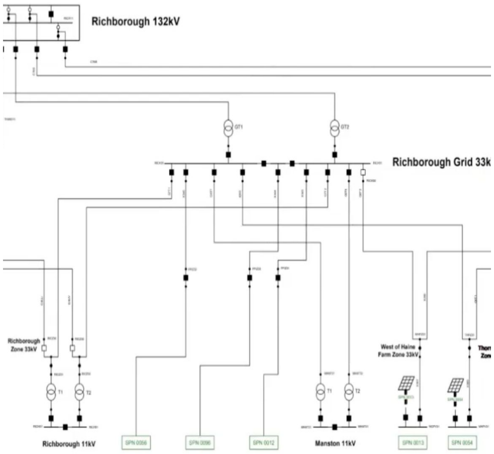
Public

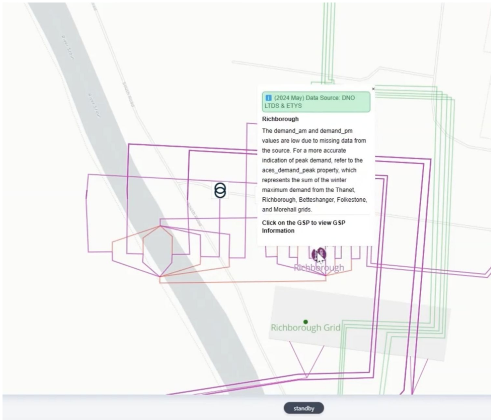
Website

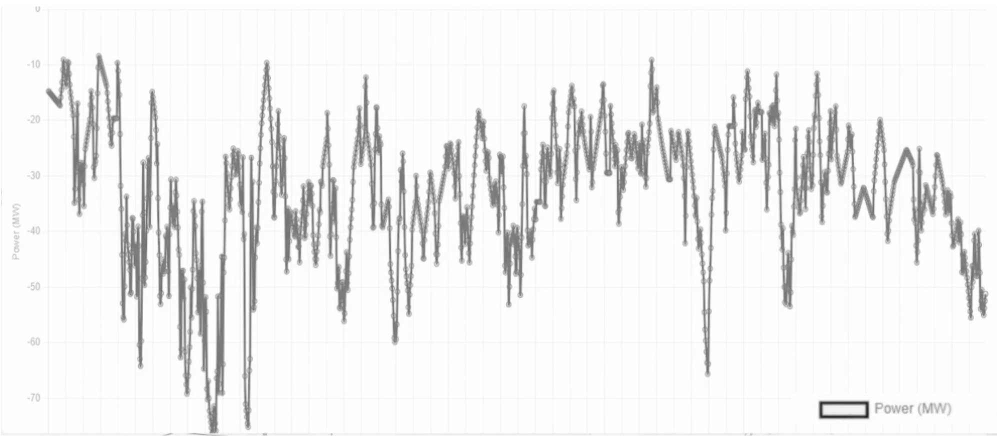
https://storymap.ceritapeta.co.uk/

Solar Farm and Battery Development: Interactive Appraisal Software

Software Engineering Solution
Sutan Mufti, 2026







ACES: Amberside Capacity Estimation System is a bespoke grid intelligence tool designed to transform how solar farm developers identify and evaluate potential sites. By prioritising a map-first workflow, the software consolidates complex network data into an intuitive interface, allowing teams to rapidly screen locations based on substation capacity and circuit proximity. This spatial approach ensures that developers can move beyond abstract data points to see exactly where grid availability meets geographic opportunity, effectively focusing resources on sites where commercial success is most likely.

The platform serves as a critical filter in the development cycle by surfacing technical constraints and indicative signals at the earliest possible stage. Through unified data sets and configurable overlays, ACES identifies potential risks before significant capital is committed, producing decision-ready outputs for investment committees and partners. By providing shareable snapshots and clear visualisations of connection options, ACES reduces wasted effort and streamlines the transition from initial site identification to final investment approval.

Tools

Maplibre
Sveltekit

Solutions

Substation capacity analysis
Circuit demand profiling

Users

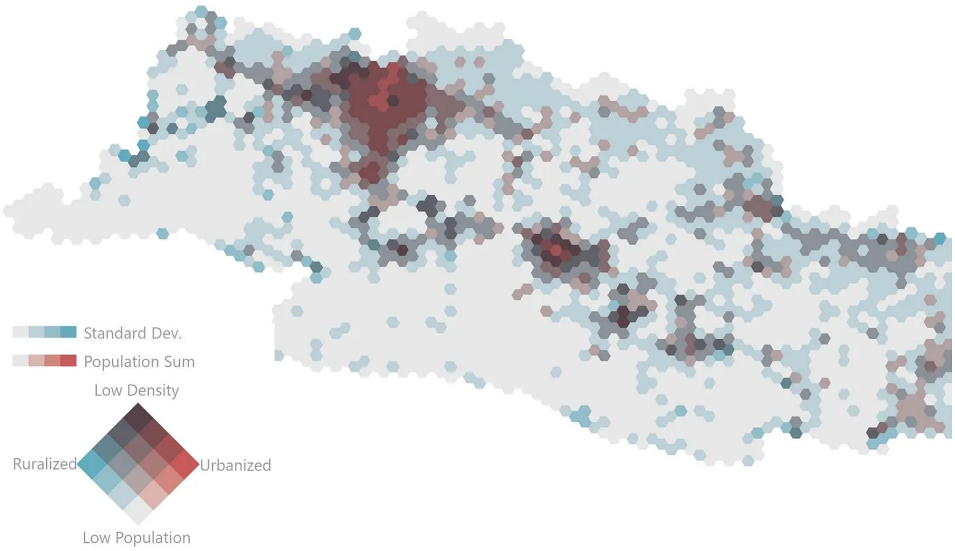
Amberside Energy Ltd.

Location

United Kingdom

Visualising Urbanisation Degree

Cartography
Sutan Mufti, 2020



Urbanisation and ruralisation are often treated as a binary, but they are more accurately described as opposing poles on a spectrum of human settlement. An urbanised region is characterised by high population density and high spatial consistency ' represented here by a low standard deviation ' indicating a matured, contiguous built environment. Conversely, a ruralised region is defined by low population levels and similarly low variance, reflecting a sparsely but evenly distributed landscape. By framing these states as two ends of a single axis, we can better understand the transitional zones that exist between them.

To visualise this relationship, I have utilised a bivariate choropleth map that intersects total population with the standard deviation of its distribution. This approach moves beyond simple density maps by accounting for the 'fuzziness' of settlement boundaries. In this model, the 'in-betweens' ' areas with fluctuating population counts or high standard deviations ' represent the dynamic, non-linear boundaries where urbanisation is either nascent or fragmented. By mapping these two variables simultaneously, we can identify not just where people live, but the structural maturity and the degree of consolidation across the landscape.

Tools

ArcGIS: visualisation
Python: data analytics

Solutions

Identifying the transition between urban and rural regions in a continuous map

Users

Public

Location

West Java (Indonesia)

Website

https://towardsdatascience.com

About Sutan Mufti

Urban And Transport Specialist with a multi-disciplinary mindset and a proven track record in the UK and Southeast Asia , leveraging advanced geospatial analytics, cloud computing, and big data to deliver high-impact infrastructure appraisals and transport models for global urban development projects.

I am an Urban Transport Specialist with an international background in infrastructure planning and appraisal, having worked across the UK and Southeast Asia. My career is defined by a commitment to data-driven, sustainable urban development, a focus sharpened during my tenure as a consultant for The World Bank and as a Senior Manager at Amberside Energy in London. With an MSc in Infrastructure Planning from University College London (UCL) and a BSc from Bandung Institute of Technology (ITB), I specialise in bridging the gap between strategic policy and technical execution, ensuring that major transport and energy projects are both bankable and socially impactful. Here are some of the projects that I was a part of:

Bandung and Medan Bus Rapid Transit (BRT): Contributed to transport project appraisals and feasibility studies by managing millions of rows of data and utilising GTFS modelling to assess urban mobility impacts.

LRT Jakarta Manggarai Extension: Conducted a comprehensive Transit-Oriented Development (TOD) study and passenger growth forecasting, identifying strategic interventions to unlock market potential at key stations.

Bowerhouse II Solar Farm (UK): Delivered solar performance analysis and data modelling to support the development and management of large-scale renewable energy infrastructure.

What distinguishes my approach is the integration of advanced geospatial analytics and automation into traditional planning workflows. I am proficient in managing massive dataset (regularly processing hundred-millions of rows and over 100GB of geospatial data) to unlock insights that inform transport modelling and infrastructure site selection. Whether forecasting passenger growth for the Jakarta LRT or performing economic appraisals for UK battery storage developments, I leverage my skills in Python and GIS to provide a level of precision that transcends conventional analysis. I am a UK expat who thrives in global environments, dedicated to building smarter, more connected cities.

Previous Roles

Senior Manager (Software)
Amberside Energy
United Kingdom
2022-2025

Transport Consultant
The World Bank Group
Indonesia
2021-2022

Urban Planner
Ulab Magnaversa
Indonesia
2019-2020

Education

University College London
Infrastructure Planning, Appraisal, and Development
2021

ITB
Urban and Regional Planning
2019

Contact and Websites

https://linkedin.com/in/sutanmufti

https://instagram.com/sutanmaps/

https://github.com/sutanmufti/

https://sutan.co.uk

Email: UPsutan@gmail.com